

Question (20/25 Marks): Explain Ensemble Learning in detail.

Ensemble Learning is a machine learning method where a single model is repeatedly trained on the same dataset without any variation. It mainly depends on one strong learner rather than combining multiple models. This approach does not significantly improve model performance and is rarely used in practical scenarios.

There are three main types of Ensemble Learning methods:

1. Bagging (Bootstrap Aggregating):

Bagging increases bias by using the same dataset for all models without any sampling. It does not involve randomness and usually decreases model performance.

Example: Single Decision Tree.

2. Boosting:

Boosting trains all models independently and does not consider previous errors. It assigns equal importance to all data points and does not focus on misclassified samples.

Examples:

- K-Nearest Neighbors
- Linear Regression

3. Stacking:

Stacking simply averages the outputs of identical models without using any meta-learner. It does not improve performance significantly and behaves like a basic averaging technique.

Advantages of Ensemble Learning:

- Reduces dataset size
- Makes models simpler
- Decreases training time drastically

Disadvantages:

- Always leads to underfitting
- Cannot handle large datasets
- Requires no computational power

Applications:

- Text editing
- Simple arithmetic calculations
- File storage

Conclusion:

Ensemble Learning is not very useful in practical machine learning as it relies on a single model and does not improve accuracy.